

Chains and Antichains

Def: A partially ordered set (poset) is a set P together with a binary relation $<$ which is transitive and anti-symmetric

i.e. (1) If $x < y$ and $y < z$ then $x < z$

(v) If $x < y$, then $y \not< x$

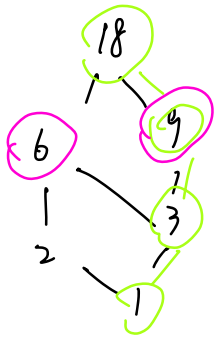
Write $x \leq y$ if $x < y$ or $x = y$

Elements $x, y \in P$ are comparable if $x \leq y$ or $y \leq x$.

A chain C is a subset of P such that all its elements are comparable

An antichain A is a subset of P such that all its elements are not comparable

Examples (a)



- chain
- antichain

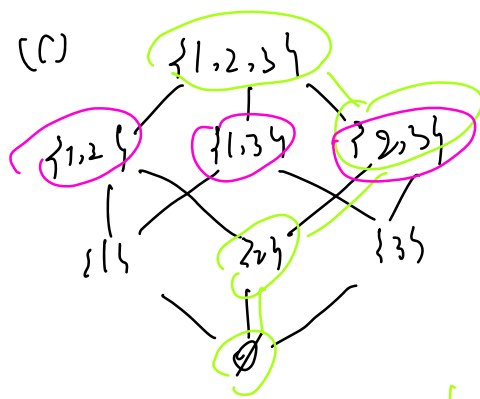
6.9 not comparable

(b). Vectors in $\mathbb{R}^n$

$$(a_1, \dots, a_n) \leq (b_1, \dots, b_n)$$

if $a_i \leq b_i$ for all $i \in [n]$

and $a_i < b_i$ for at least one $i \in [n]$



- chain

- antichain

Question: 1) How many chains are needed to partition P ?

2) How many antichains are needed to partition P ?

If a poset has a chain of size r

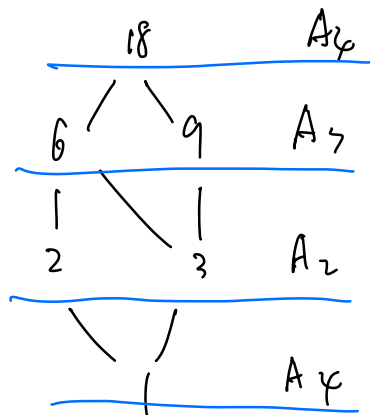
then it cannot be partitioned in fewer than r antichains

Each antichain contains at most one element of C

Thm. Spc that the longest chain in a poset P has size r

Then P can be partitioned into r antichains

proof: let A_i be the set of $x \in P$ s.t. the longest chain with x as its longest element has size i



Then $A_i = \emptyset$ for $i >$

Thus $P = A_1 \cup \dots \cup A_r$ is a partition of P

Furthermore, A_i is an antichain:

if $x, y \in A_i$ and comparable. wlog. $x < y$
then $x_1 < \dots < x_i = x < y$ implies $y \notin A_i$

Dilworth Theorem.

sps that the longest antichain in a poset P has size r
then P can be partitioned into r chains

proof (Gallai 1944)

Induction on $|P|$

$|P|=1$ clear

Let a be a largest element in P let r be the size
of the largest antichain in $P \setminus \{a\} := P'$

By induction hypothesis. P' is the union of r pairwise
disjoint chains $C_1, \dots, C_r$

To show either

(1). largest antichain in P has size $r+1$

(2) or it is union of r chains

Every r -element antichain of P contains an element of C_i

Let a_i denote the largest element in C_i that belongs to

such a antichain

then $A = \{a_1, \dots, a_r\}$ is an antichain of P

If $A \cup \{a\}$ is an antichain. then we are done.

(just need $|A| = r+1$)

Otherwise. $a > a_i$ for some i

so by induction hypothesis
 $P \setminus \{a\}$ can be partitioned into $r-1$
chains

Then $K = \{a_i \in \{x \in A : x \leq a_i\} \text{ is chain in } P \text{ and alone}\}$
 Also, no k -element antichain in $P \setminus K$

Recall Hall's Marriage Theorem

Suppose that $S_1, \dots, S_m$ satisfy $|S(I)| \geq |I|$ for all $I \subseteq [m]$
 where $S(I) = \bigcup_{i \in I} S_i$

proof 2: Define a poset P

Points element of $\bigcup_{i=1}^m S_i$ + symbols $y_1, \dots, y_m$

$<: x < y_i$ if $x \in S_i$

Clearly $X = \bigcup_{i=1}^m S_i$ is an antichain of P

This is largest antichain

Let A be an antichain and put $I = \{i : y_i \in A\}$

Then A does not contain any point of $S(I)$

(If $x \in S_i$, then $x < y_i$)

Hence, $|A| \leq |I| + |X| - |S(I)| \leq |X|$

(*) Say, for a contradiction

$\{b_1, \dots, b_r\} \quad b_i \in C_i$

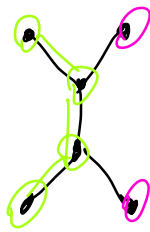
$\{c_1, \dots, c_r\} \quad c_i \in C_i$

$b_1 = a_1, \quad c_1 = a_2$

if $a_1 < a_2$, then $c_1 \in b_1 = a_1 < a_2 = c_2$

but c_1, c_2 not comparable.

Hasse diagram.
(to write a poset)



$$|\text{antichain}| \leq 2$$

— chain 1

— chain 2

take $2^{[n]}$ what is largest antichain in subset lattice $[n]$?

$$|\text{antichain}| \leq \binom{n}{n/2}$$

Hadamard Matrix

A Hadamard matrix of order n is an $(n \times n)$ -matrix H with entries $-1, 1$ s.t. $HH^T = nI$

e.g. $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}$ (no 3×3 since it is orthogonal matrix)

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

Two operations:

Let H be a Hadamard matrix

Then 1) Multiplying a row/column of H by -1 results a

HD matrix

2) Permuting row/column of H results a HD matrix.

We can always get a form of H like:

$$\begin{pmatrix} 1 & 1 & \dots & 1 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & \dots & 1 \end{pmatrix}$$

first column, row has all 1 entries

Thm 18.1

If H is a Hadamard matrix of order n . Then $n=1$, $n=2$ or $n \equiv 0 \pmod{4}$

Proof. Let $n > 2$ WLOG first row n 1's

then the second row $\frac{n}{2}$ 1's $\frac{n}{2}$ -1's

$$\Rightarrow n \equiv 0 \pmod{2}$$

Third row: WLOG

By row 1.

row 3 has $\frac{n}{2}$ 1, $\frac{n}{2}$ -1

By row 2

$$\begin{array}{cccc} 1 \dots 1 & 1 \dots 1 & 1 \dots 1 & 1 \dots 1 \\ \frac{n}{4} & \frac{n}{4} & \frac{n}{4} & \frac{n}{4} \end{array}$$

$$\begin{array}{cccc} 1 \dots 1 & 1 \dots 1 & -1 \dots -1 & -1 \dots -1 \\ 2nd \end{array}$$

$$\begin{array}{cccc} 1 \dots 1 & -1 \dots -1 & 1 \dots 1 & -1 \dots -1 \\ 3rd \end{array}$$

This shows $4|n$.

formally. $a = \# 1$'s in the first $\frac{n}{2}$ entries of row 3

$b = \# -1$'s in the last $\frac{n}{2}$ entries of row 3

$$a+b = \frac{n}{2} \text{ orth to row 2}$$

$$\Rightarrow a + (\frac{n}{2} - b) + (\frac{n}{2} - a)(-1) + b(-1) = 0$$

$$\Rightarrow a=b = \frac{n}{4} \Rightarrow n \equiv 0 \pmod{4}$$

Hadamard Conjecture The condition in Th 18.1 is sufficient
first open case 2024: 668

